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Paleomagnetism of Carboniferous Intrusions in North Carolina

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Paleomagnetic samples of two late Paleozoic plutons were collected from the eastern Piedmont Province in North Carolina. Results from the early Carboniferous (326 ± 27 m.y.) Lilesville pluton, located in the Carolina Slate Belt, indicate a pole position similar to that of the stable North American craton for the lower Carboniferous. Five sites in this gabbro/granite complex give a pole of 40°N, 134°E ($\alpha_{95} = 2.9$). The late Carboniferous 282 ± 6 m.y.) Churchland pluton, located in the Charlotte Belt, gives a pole of 34°N, 126°E ($\alpha_{95} = 16.3$) which correlates well with other North American poles for the late Carboniferous. Observations made by Kent and Opdyke (1978) on rocks from the eastern edge of the northern Appalachians suggest gross lateral displacement between the mid-Devonian and late Carboniferous in the late Paleozoic. This paleomagnetic information from the eastern Piedmont region implies that the area of the Carolina Slate Belt and Charlotte Belt was associated with the North American platform for the entire Carboniferous period. The Slate Belt rocks, although geologically a southern counterpart to some of the Avalonian rocks from the northeastern edge of North America, do not display a paleomagnetic history suggestive of great displacement.

INTRODUCTION

The geology of the southern Appalachians consists of a series of NE-SW trending belts that define distinct geologic terrains (Figure 1). The evolutionary history of these belts and their relationship to each other and to Appalachian plate tectonics remains uncertain. Information obtained from recent studies of the eastern margin of North America has had a major impact on the interpretation of southern Appalachian geology.

Paleomagnetic investigations of the eastern margin of North America indicate that the Canadian Maritimes-eastern New England region may not have been a contiguous part of the North American continent prior to the late Carboniferous [*Kent and Opdyke*, 1978]. These authors explain the relative southerly displacement of this region by suggesting that left-lateral movement brought these areas into their present juxtaposition with the stable North American platform. It is not known if the southeastern edge of the continent should be incorporated into this theory. The surface expression of a suture that would substantiate this hypothesis has not been located.

A seismic travese through the southern Appalachian Piedmont has revealed a lower Paleozoic sedimentary layer below the precambrian to Paleozoic rocks of the eastern part of the orogen [*Cook et al.*, 1979]. Their interpretation indicates a possible 260-km thrust of the eastern belts westward over the Paleozoic eastern continental margin of North America.

Investigations of the complex geology of the eastern margin of North America have provided substantial evidence for an ancient (600 m.y.) 'Avalonian microcontinent' that was welded to the main mass of North America in a later collisional event. The total extent of the Avalon block and the precise timing of its addition to the North American craton is poorly known [*Brown*, 1979; *Brown and Kelly*, 1980; *Van der Voo et al.*, 1979]. The age and origin of the crust below the eastern belts of the Piedmont has not yet been determined. The role the Avalon block played in the evolution of this area is still unknown.

Petrologic studies of mafic plutonic rocks in the southern Appalachians imply that the Carolina Slate Belt originated as an island arc [*Butler and Fullagar*, 1978; *Whitney et al.*, 1978]. The low initial 87 Sr/86 Sr ratios detected in several late Paleozoic grantic plutons of the eastern Piedmont, thought to be the result of contamination by partial melting of mafic lower crust, support the island arc model. This interpretation does not permit a Grenville age for the basement rock of the eastern Piedmont. Alternatively, heat flow measurements of Piedmont granites correlate with continental crust beneath the Slate Belt of a Grenville or older age [*Glover et al.*, 1978; *Costain and Glover*, 1980].

This paleomagnetic study of two late Paleozoic plutons of the Carolina Slate Belt and Charlotte Belt region was undertaken to contribute information about the eastern Piedmont region to the available Paleozoic paleomagnetic data. The area was selected as an appropriate site to test current interpretations of the geology of the eastern margin of North America.

TECTONIC SETTING

Regional geology. The Carboniferous geology of the eastern margin of North America suggests continent-continent collision [*Hatcher*, 1978]. Evidence of plutonism associated with the subducted continental crust is apparent in the large number of Carboniferous age intrusive bodies that have been located in the eastern Piedmont region (Figure 1). The postmetamorphic plutons that define two parallel linear trends (a major eastern and smaller western group) along the southeastern margin of North America have been examined in detail structurally and petrographically [*Speer et al.*, 1980]. These plutons appear to be a supergroup similar in age and petrographic character; a common origin of these plutons is suspected. The linearity of the belts of granite plutons suggests generation of magma above a subduction zone [*Fullagar and*

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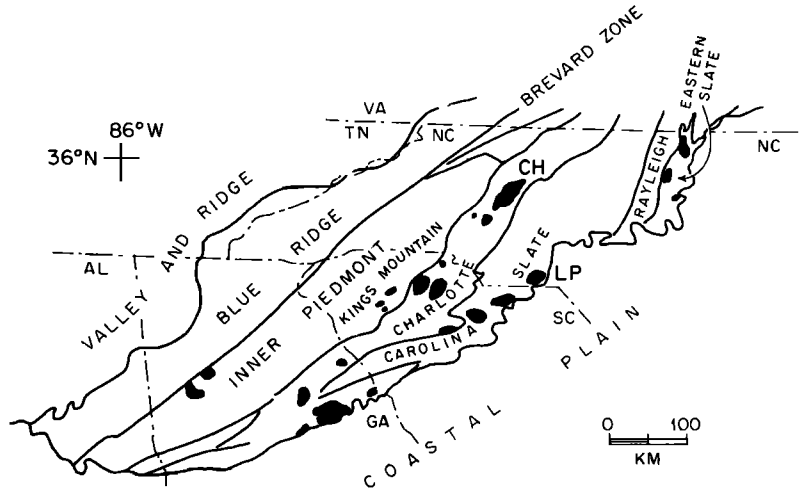


Fig 1. Setting of southern Appalachians, showing tectonic belts. Post-metamorphic granitoid bodies shown in black [after *Speer et al.*, 1980]. LP, Lilesville pluton; CH, Churchland granite.

Butler, 1979]. Geochemical investigation of the eastern Piedmont plutons detected similarities in the oxygen isotope composition and trace elements along strike of the linear trend and systematic variations across strike [*Sinha and Zietz*, 1980].

The lack of widespread compressional deformation in the eastern Piedmont in the late Paleozoic has been well documented. The 325-265 m.y. plutons show no sign of deformation microscopically or in field relations [*Fullagar and Butler*, 1979]. To the west in the Blue Ridge and western Piedmont there is abundant evidence for deformation and metamorphism at this time [*Kish et al.*, 1979; *Tull*, 1980]. If a Pennsylvanian to Permian collision with Africa was the cause of the deformation to the west, a corresponding deformation would be expected in the east. This striking discrepancy in the Carboniferous geology of the southern Appalachians has led to several interpretations. A major fault system has been suggested for the eastern Piedmont to account for the necessary crustal adjustments in the late Paleozoic central and eastern Piedmont [*Hatcher et al.*, 1977]. Results of detailed seismic profiling support the theory of a major sole thrust upon which the Blue Ridge, Inner Piedmont, and eastern belts were transported westward a distance of approximately 260 km [*Cook et al.*, 1979].

An alternative view, originally proposed by *Dennison* [1976], which accounts for the lack of deformation in the Pennsylvanian to Permian plutons is that collision produced the uplift of the Piedmont. Gravitational spreading caused by this uplift resulted in the late Paleozoic thrust faulting and folding in the Valley and Ridge and Blue Ridge region. A vast accumulation of Pennsylvanian to Permian sediments in the Appalachian basin from a source to the east supports Dennison's theory. Geochronologic evidence from the eastern Piedmont favors the uplift, gravity spreading model [*Long et al.*, 1959; *Fullagar*, 1971].

Pee Dee Lilesville complex. The Pee Dee gabbro and Lilesville granite crop out in the eastern Carolina Slate Belt. The Slate Belt is composed of late Precambrian to early Paleozoic volcanogenic rocks that have been metamorphosed to the greenschist facies and intruded by a variety of plutonic rocks. One of these plutons is the Lilesville, a concordant sheet structure with an areal extent of 325 km^2 and a maximum thickness of 2.8 km [*Waskom and Butler*, 1971] (Figure 2). The large batholith is compositionally zoned adamellite, granodiorite, and tonalite. The Pee Dee gabbro, the focus of the paleomagnetic work, intrudes the eastern portion of the Lilesville granite. The areal extent of the gabbro is 15 km^2. Part of the Lilesville pluton is cut by the border fault of the

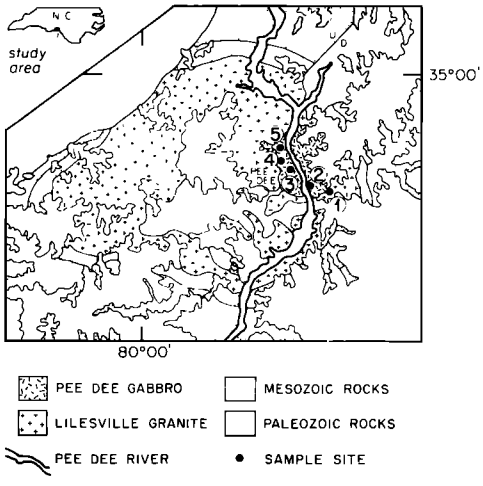


Fig. 2. General geology, structure, and sample sites of the Lilesville pluton, granite, North Carolina [after *Waskom and Bulter*, 1971].

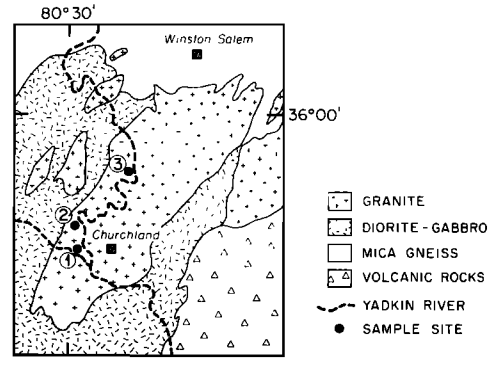


Fig. 3. General geology and sample sites of the Churchland granite, North Carolina [after *Sanders*, 1958].

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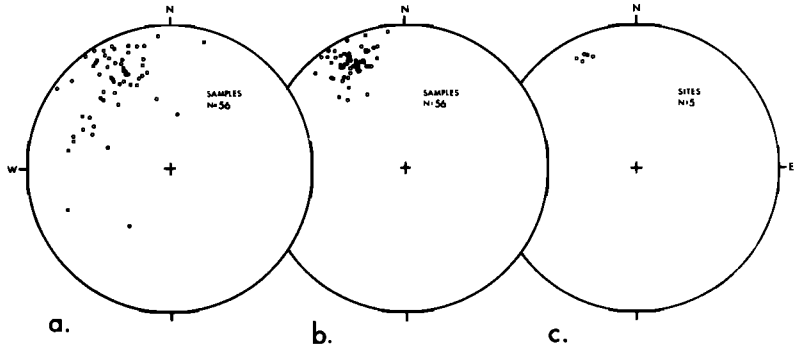


Fig. 4. Equal-area projections of magnetization directions from the Lilesville pluton. Open circles are on the upper hemisphere; solid circles are on the lower hemisphere. (*a*) Natural remanent magnetization, all samples. (*b*) magnetiztion direction after demagnetization at 10 mT, all samples. (*c*) Magnetization directions after demagnetization at 10 mT, all sites.

Wadesboro and Ellerbe Triassic basin, but it is believed to be undeformed except along the fault boundary [*Waskom and Butler,* 1971].

The plutons in this part of the Piedmont were emplaced after the late Paleozoic metamorphism and penetrative regional deformation but before the intrusion of diabase dikes, the uplift, and the erosion that occured during the Mesozoic [*Butler,* 1966; *Butler and Ragland,* 1969; *Jones and Walker,* 1973; *Bell et al.*, 1974; *Speer et al.*, 1980]. The Lilesville pluton has been dated by *Fullagar and Butler* [1979] at 325 27 m.y. Examination of the petrographic relationship between the gabbro and granite facies of the pluton shows that the granite had not completely cooled at the time of the gabbro intrusion [*Speer et al.*, 1980]. Fragments of fine-grained 'pillowlike' inclusions of gabbro in the granite point to a contemporaneous intrusion of the two magmas. Although no specific age date is available for the Pee Dee gabbro, its magmatic relationship to the Lilesville granite supports a similar early Carbonferous age.

Churchland pluton. The Churchland granite is a late Paleozoic pluton located in the northern part of the Charlotte Belt of the Appalachian Piedmont (Figure 3). The Charlotte Belt has a wide compositional range of intrusive rocks and some high rank metamorphic rocks. The characteristic meta-sedimentary and meta-igneous rocks of the Charlotte Belt are intruded by both premetamorphic and postmetamorphic plutons. The Churchland is one of the postmetamorphic plutons and believed to be part of the same magma series that produced the Lilesville, Pageland, Clouds Creek, and several other postmetamorphic granitic plutons in the southern Appalachians [*Fullagar and Butler,* 1979]. The Churchland belongs to a western group of plutons that is distinguished from an eastern group on the basis of western location and a higher Sr content. Again, the Churchland pluton is clearly younger than the regional metamorphism of the area. Petrographic studies by *Privett* [1973] on the Churchland confirm that it is undeformed. Based on K/Ar analysis of whole rock and biotite samples, the age of the Churchland is 282 ± 6 m.y. [*Fullagar and Butler,* 1979].

Paleomagnetic Studies

A portable gasoline-powered rock drill was used for the collection of paleomagnetic samples. The 2.5-cm-diameter cores were drilled in well-exposed outcrops where there was little sign of weathering. Spacing between cores was usually 50 cm, and cores were drilled to a depth of 5-10 cm.

All cores were cut to 2.5-cm lengths with a rock saw. Usually, only one specimen was obtained per core. The natural remanent magnetization (NRM) of each sample was measured on a Princeton Applied Research (PAR) spinner magnetometer. In order to isolate the characteristic magnetic directions, alternating field (af) demagnetization experiments were performed on representative samples from each site of the bodies sampled using a Schonstedt demagnetizer. Specimens were subject to progressively higher alternating fields in seven steps from 2.5 to 100 mT and remeasured after each treatment. Thermal demagnetization of the Churchland samples was done in a shielded oven and cooled in a field of less than 50 nT.

Results

Pee Dee/Lilesville complex. A total of 56 oriented samples were drilled at five sites in the Pee Dee gabbro and Lilesville granite. Four of the five sites are in the more mafic gabbro body, and the site in the Lilesville granite is close to the granite-gabbro contact (Figure 2). The NRM of the gabbro is fairly strong, averaging 0.55 A/m. The granite is relatively weak with an average intensity of 0.12×10^{-2} A/m. The NRM directions for all samples show some scatter but are dominantly pointing up and to the northwest (Figure 4*a*).

The change in remanent intensity with higher alternating fields is uniform for all of the progressively demagnetized samples. There is a decrease to one half the initial value between 10 and 20 mT for both the granite and gabbro samples. The remanent intensity of the gabbro is reduced to one third its initial value between 20 and 40 mT, and the remanent intensity of the granite drops to one third its initial value between 40 and 80 mT. In samples from the four gabbro sites the decrease in intensity is accompanied by very little change in remanent direction (sites 3 and 5), whereas the remanent directions of the granite samples (site 1) vary widely before reaching a stable direction.

The granite samples change progressively from a moderate inclination and westerly direction to a more shallow inclination and northwesterly direction. Two orthogonal projections of typical demagnetization trajectories at each demagnetization

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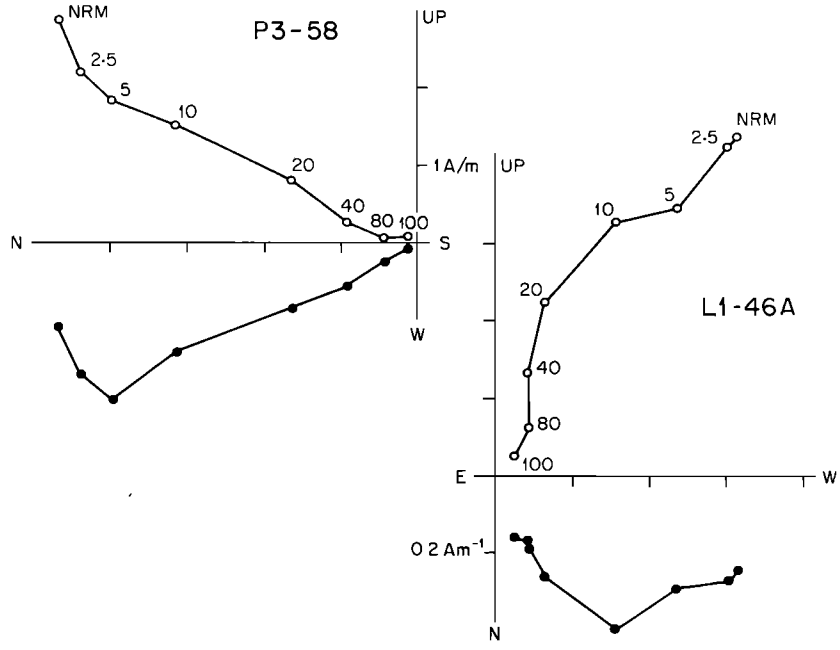


Fig. 5. Orthogonal alternating field demagnetization diagrams for representative samples from the Lilesville pluton. Open circles are projections on the vertical plane; solid circles are projections on the horizontal plane. Field strength is in millitesla. Sample P3-58 is from the Pee Dee gabbro; sample L1-46A is from the Lilesville granite.

level are shown in Figure 5. All gabbro samples exhibit a stable remanent component after demagnetization to 10 mT. A steep northerly viscous component, evident in many samples, is removed at low demagnetizations levels (e.g., P3-58, Figure 5). The Lilesville granite samples do not reach a stable direction until at least the 20mT level (e.g., L-46A, Figure 5). The levels of cleaning for the gabbro and granite samples were 10 and 20 mT, respectively. Characteristic directions of all samples (Figure 4*b*) and each site (Figure 4*c*) are well clustered after cleaning.

The acquisition of the thermoremanent magnetic direction for the Lilesville pluton appears to have occurred upon cooling of the body. The stability of all samples through af demagnetization indicates only one high coercivity component, assumed to be the primary remanence. Table 1 summarizes the mean directions and statistical parameters for each site in the Pee Dee/Lilesville complex. The site mean directions have an excellent cluster giving a direction with declination 334.4°, inclination -16.3°, $k = 41.4$ and $\alpha 95 = 2.9$. The pole position obtained from the average of all samples is 40°N, 134°E.

Churchland pluton. Reconnaissance sampling of the Churchland granite was in the area of Cooleemee and Churchland, North Carolina, near the Yadkin River (Figure 3). Thirty-one orientated samples were collected from three sites in the Churchland granite. The NRM of the granite is extremely variable in intensity and direction, with an intensity range of 0.11 A/m to 0.50 x 10^{-3} A/m. The NRM directions show substantial scatter but a preference toward the southwest with positive or shallow negative inclinations (Figure 6*a*).

All samples were remeasured after being subjected to 10, 20, 30, and 40 mT fields of demagnetization. The change in remanent intensity with higher alternating fields is similar in most of the Churchland granite samples. Over half of the original

TABLE 1. Paleomagnetic Directions and Pole Positions

Site	N	H	I	D	k	α95	Pole	Delp	Delm
Lilesville Pluton									
1	10	10	-15.5	336.9	16	12.3	42°N, 132°E	6.5	12.7
2	7	10	-14.9	330.7	33	10.7	39°N, 140°E	5.6	11.0
3	19	10	-18.0	332.1	78	3.8	39°N, 136°E	2.1	4.0
4	10	10	-18.7	338.4	55	6.6	41°N, 129°E	3.6	6.8
5	10	10	-12.7	334.6	89	5.2	42°N, 135°E	2.7	5.3
Mean	56	10	-16.3	334.3	41	3.0	40°N, 134°E	1.6	3.1
Churchland Pluton									
1	11	40/400	-44.7	327.2	17	11.5	20°N, 131°E	9.1	14.5
2	8	40/400	-37.6	314.6	5	29.4	19°N, 144°E	20.4	34.6
3	10	40/400	+6.1	9.9	3	34.0	56°N, 82°E	17.1	34.1
Mean	29	40/400	-29.8	337.4	4	16.3	34°N, 126°E	10.0	18.1

N is a number of samples per site; *H* is demagnetization level, in mT or degrees centigrade; *I* and *D* are magnetic directions, in degrees; *K* is the precision parameter; α95 is the radius of the cone of 95% confidence about the mean; Delp and Delm are the minor and major axes of the oval of confidence about the pole.

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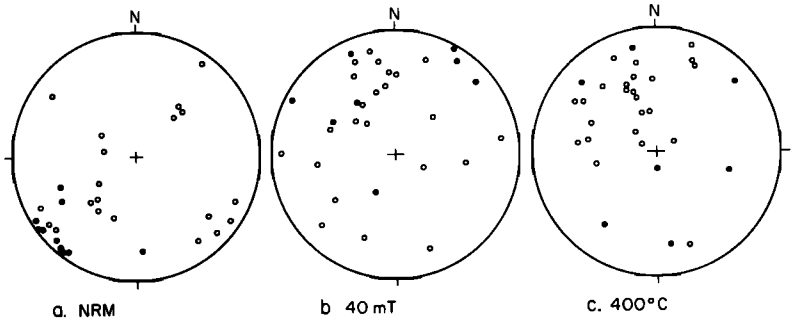


Fig. 6. Equal-area projections of magnetization directions from the Churchland granite. Symbols as in Figure 4. (*a*) Natural remanent magnetization directions. (*b*) Directions after alternating field demagnetization to 40 mT. (*c*) magnetization directions after addition thermal demagnetization to 400°C.

intensity is lost by 10 mT in a majority of the samples. Only one third of the remanent intensity remains after demagnetization in excess of 20 mT in all specimens. The decrease in intensity is usually accompanied by extreme changes in remanent direction. Figure 6*b* is a plot of magnetic directions for all samples after 40 mT cleaning. The southwesterly component has been removed and although a northerly component is indicated, within site precision is very poor. Low coercity components could not be isolated and *Zijderveld* [1967] plots (Figure 7) reveal complex demagnetization curves until the 40-mT level.

In an effort to reduce direction scatter, thermal demagnetization experiments were conducted on several samples of the Churchland granite. Samples were remeasured at 100°C intervals from 150° to 450°C. Thermal demagnetization was concluded after the 450°C experiment due to the decomposition of the granite at this temperature. Figure 8 is a plot of typical sample directions from sites 1 and 3 with successive alternating field demagnetization followed by thermal demagnetization. Many of the scattered samples move into a northerly position with thermal demagnization (e.g., sample 3-183, Figure 8). Samples that had secured a northerly position with af treatment maintain that position at higher temperatures (e.g., sample 1-164, Figure 8).

A temperature of 400°C was selected as an optimal treatment level for the Churchland granite. The results of the final thermal cleaning are shown in Figure 6*c*. Five samples, all from site 2 have directions 180° from the majority of the samples. These reversed directions may imply a change in the earth's magnetic polarity over the cooling history of this large (400 km^2) body. This assumption is considered valid for the extensive Churchland pluton, and these few samples were reversed in direction before they were incorporated into the final statistical analysis.

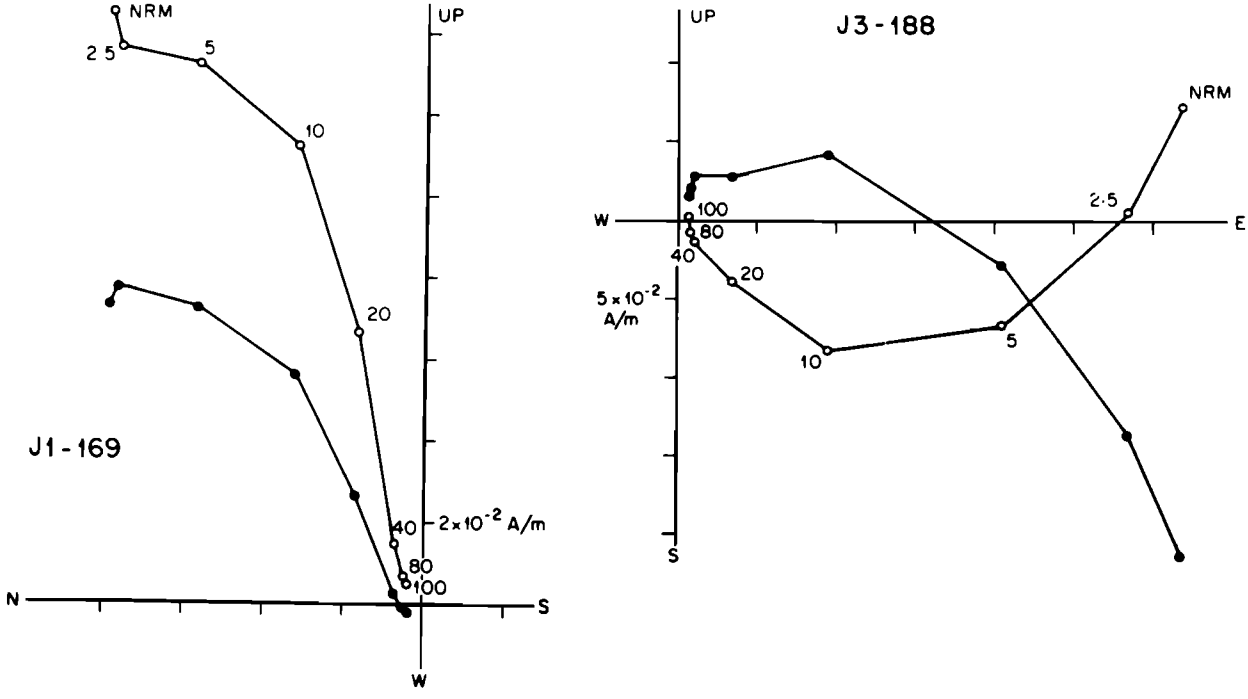


Fig. 7. Orthogonal alternating field demagnetization diagrams for representative samples from the Churchland granite. Symbols as in Figure 5.

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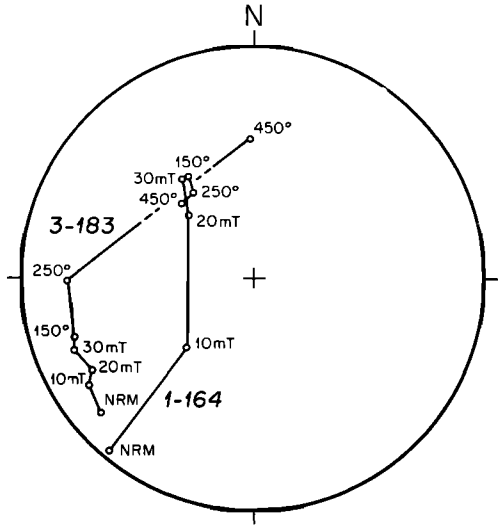


Fig. 8 Equal-area plot of Churchland granite samples, showing change in magnetization directions after progressive alternating field demagnetization followed by thermal demagnetization.

Mean directions for the three Churchland sites and the corresponding statistical parameters are listed in Table 1. The within site precision parameter, k, is low for all sites. As reported, several samples behaved erratically with successive cleaning in alternating fields of 10, 20, 30, and 40 mT. Within site precision improved very slightly with thermal demagnetization, from 1.7 to 3.4 on the average. The migration toward a northerly component apparent at the 40-mT level becomes more obvious at 400°C.

Similar erratic behavior on af demagnetization has been noted in weathered samples of the Milledgeville granite [*Ellwood,* 1981]. Comparison of the weathered samples to unweathered cores indicated that the directional scatter was caused by partial maghemitization of magnetite grains. Ellwood was able to obtain the same stable remanent direction from both sets of samples after extensive demagnetization.

Pole positions and site mean directions calculated from the Churchland sample data after thermal cleaning are dissimilar, and for this reason the interpretation of these results must be accompanied by a measure of speculation. The mean direction obtained from the 29 samples for the Churchland granite is declination 337.4, inclination -29.8, $k=4$, and $a95=16.3°$. The paleopole of the Churchland granite falls at 34°N, 126E.

Magnetic Mineralogy

Pee Dee/Lilesville complex. Reflection microscopy of selected samples of the Pee Dee gabbro and Lilesville granite was done to determine the oxide minerals carrying the characteristic magnetism of this body. The gabbro has an abundance of oxides that have undergone a minimal amount of oxidation. Magnetite grains are commonly adhedral and 0.7 mm in size; a fine dust of 0.01-mm magnetite grains throughout the gabbro with no preferred orientation is also noted. Adhedral to subhedral ilmenite grains are present, commonly over 0.7 mm in size, with fine hematite lamellae that are less than 0.001 mm long. The Lilesville granite contains unaltered homogeneous magnetite. Grains are 0.1 mm, subhedral and show no oxidation. Large, unaltered grains of ilminite are observed, most are discontinuous but in optical continuity.

Churchland pluton. The most abundant magnetic mineral in the Churchland granite is magnetite. The magnetite grains range between 0.1 and 0.7 mm, and all have been altered to some extent to maghemite with no secondary hematite formation. Maghemitization is a low-temperature event that can occur during the late stages of initial cooling, or can be diagnostic of supergene weathering since it can only form naturally by oxidation at low pressure [*Haggerty,* 1976]. The erratic magnetic behavior and physical disintegration of samples upon heating indicate that the Churchland has undergone considerable weathering. Relect ilmenite is also present in the Churchland granite.

Discussion

The pole of 40°N, 134°E obtained from the Pee Dee/Lilesville study agrees well with three other reliable lower Carboniferous studies, as seen in Figure 9 and Table 2. Samples of the Mauch Chunk siltstone of the folded Appalachians in Pennsylvania give a pole at 43°N, 127°E [*Knowles and Opdyke,* 1968]. The Barnett shale of Texas has a lower Carboniferous pole at 49°N, 119°E [*Kent and Opdyke,* 1979]. The Saint Joe limestone in Arkansas gives a pole of 39°N, 132°E after thermal cleaning [*Scott,* 1979]. Each of these studies provides a paleopole for the stable North American craton in the lower Carboniferous.

The pole position of 34°N, 126°E calculated from the Churchland results lies close to the results from nearly all upper Carboniferous paleomagnetic studies (Figure 9 and Table 2). There are five paleomagnetic studies that can be used to determine the upper Carboniferous position for the North American craton. Results from Ohio, West Virginia, Kentucky, Pennsylvania, Colorado, and Wyoming combined give an average pole at 37°N, 128°E, whereas studies from the Canadian Maritimes (Nova Scotia, New Brunswick, and Prince Edward Island) average to a pole position of 39°N, 129°E. The agreement between all upper Carboniferous studies indicates that the eastern margin of North America had assumed its present configuration by the upper Carboniferous.

The Pennsylvania Churchland and the Mississippian age Pee

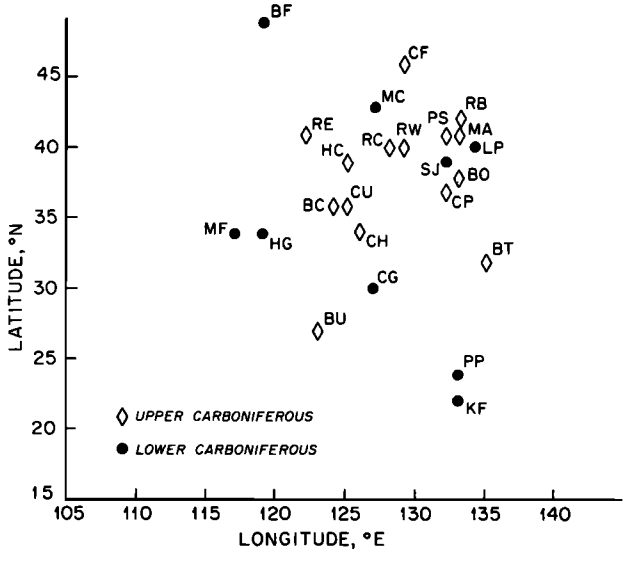


Fig. 9 Plot of Carboniferous north poles for North America - study identification, pole position, and source given in Table 2.

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TABLE 2. Carboniferous Paleomagnetic Pole Positions, North America

Unit	*Key**	*Pole*	*Reference***
Upper Carboniferous			
Bathurst Formation, N.B.	BT	32°N, 135°E	*Black* [1964]
Bonaventure Formation, N.B.	BO	38°N, 133°E	8-100
Brush Creek Limestone, Pa.	BC	36°N, 124°E	*Payne et al.* [1981]
Buffalo Siltstone, Pa.	BU	27°N, 123°E	*Payne et al.* [1981]
Casper Formation, Wyo.	CF	45°N, 129°E	*Diehl and Shive* [1981]
Churchland Pluton, N.C.	CH	34°N, 126°E	this study
Coal Pole, Ohio/W. Va.	CP	37°N, 132°E	*Kopacz and Noltimier* [1976]
Cumberland Group, N.S.	CU	36°N, 125°E	11.76
Hurley Creek Formation, N.B.	HC	39°N, 125°E	9-98
Maroon Formation, Colo.	MA	41°N, 133°E	*Christensen and Helsley* [1974]
Pictou Sandstone, N.B.	PS	41°N, 132°E	8-101
Red Beds, East P.E.I.	RE	41°N, 122°E	8-95
Red Beds, Central P.E.I.	RC	40°N, 128°E	8-96
Red Beds, Northwest P.E.I.	RW	40°N, 129°E	8-97
Red Beds, P.E.I.	RB	42°N, 133°E	8-99
Lower Carboniferous			
Barnett Formation, Tex.	BS	49°N, 119°E	*Kent and Opdyke,* [1979]
Codroy Group, Nfld.	CG	30°N, 127°E	8-117
Hopewell Group, N.B.	HG	34°N, 119°E	11-78
Kenniebecasis Formation, N.B.	KF	22°N, 133°E	*Black* [1964]
Lilesville Pluton, N.C.	LP	40°N, 134°E	this study
Maringouin Formation, N.B.	MF	34°N, 117°E	10-120
Mauch Chunk, Pa.	MC	43°N, 127°E	10-119
pre-Pictou Sediment, N.B.	PP	24°N, 133°E	8-113
St. Joe Limestone, Ark.	SJ	39°N, 132°E	*Scott* [1979]

* Symbol used in Figure 9.
**Reference numbers refer to *McElhinney* [1968-1980].

Dee/Lilesville complex have pole positions that define their original location to be cratonic North America. The northward shift of the paleopole position of the Maritimes-coastal New England region between the lower and upper Carboniferous that is apparent through the analysis of the available paleomagnetic data [*Kent and Opdyke,* 1978, 1979] (Table 2) is not observed for the eastern Piedmont region.

These Carboniferous paleomagnetic results concur with an evolutionary theory of the eastern margin of North America that proposes the Carolina Slate and Charlotte belts had cratonic affinities by the lower Carboniferous. Cooling of these intrusive bodies in situ, not northward transport, is suggested by this study. Two points specific to the possible westward transport must by noted with this interpretation. The accuracy of paleomagnetic statistics is dependent upon a circle of confidence that can correspond to hundreds of kilometers of the earth's surface. Even large-scale movements along thrust faults could go undetected with a paleomagnetic approach to regional geology. The second point is the absence of longitudinal information available from paleomagnetic data. In this particular study the major movement of the continental blocks may have been roughly east-west, and the change in points of longitude for a land mass could not be obtained. Because of these constraints the supposed allochthony of the eastern Piedmont along a major thrust fault can not be determined with certainty.

The hypothesis of 1500 km of left-lateral movement was suggested by *Kent and Opdyke* [1978] to account for the discrepancy between paleomagnetic results from the craton and the northeastern margin of North America for middle and upper Paleozoic times. Paleomagnetic data show the late Carboniferous as the earliest time that the North American platform and the Maritimes-eastern New England region share a common polar wander path. A recent report by *Diehl and Shive* [1981] on the Casper formation of Wyoming supports this basic premise and extends the time of coincidence of the stable craton and Maritimes-eastern New England polar wander paths into the Permian. Such an interpretation is not supported by the data as presented in Table 2.

In the foregoing discussion of the magnetic directions found preserved in the bodies studied, it is assumed that the plutons have not experienced tectonic tilting or rotation subsequent to their initial magnetization. It has long been established that the eastern margin of North America underwent a period of block faulting in the early Mesozoic. The result is a series of fault bounded basins that extend from Nova Scotia to the Carolinas [*Longwell,* 1933]. Because these basins are clearly a feature of the study area and also in light of the 20° offset in some of the paleomagnetic studies of the eastern margin of North America [*Kent and Opdyke,* 1978], the possible effect of this post-Paleozoic movement on the results from this study was explored. The Pee Dee/Lilesville complex located just east of Wadesboro Basin was selected to test the effects of an undetected 20° eastward tilt. The correction of N45°E, 20° was applied to the average inclination of -16° and declination of 334° for this body. The resulting magnetic vector has an inclination of 40° and declination of 332° and a corresponding pole of 48°N, 145°E. This hypothetical pole does not suffice to compensate for the 20° offset in latitude detected by paleomagnetic studies in the Maritimes and eastern New England region. For the situation given here, it in fact increases the offset in latitude. Given the variety of Mesozoic fault-bounded basins in the area of the Piedmont, it is possible that other directions of tilting could have occurred. Without precise structural information it cannot be ruled out that results from any intrusive body may have a tilt or rotation in-

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corporated into the resultant magnetic vector obtained from paleomagnetic study.

The effect of a Mesozoic tilt was not incorporated into the study results for two reasons. First, the calculated change in the pole latitude as a result of a structural tilt is less than the actual offset in pole latitude recorded in Martimes-eastern New England palemagnetic studies. Second, a tilt correction would need to be applied to all Palezoic paleomagnetic studies of the eastern margin of North America for a valid comparison.

Conclusions

Results from this study of the Pee Dee/Lilesville complex and Churchland granite indicate that the area of the Carolina Slate Belt and Charlotte Belt may have been an integral part of cratonic North America by the lower Carboniferous. Preliminary results from the paleomagnetic analysis of early Paleozoic sills located in the Denton-Albermarle area of the Carolina Slate Belt imply that the eastern Piedmont may have had an affinity with cratonic North America as early as mid-Ordovician time [*Barton,* 1980; *Brown and Barton,* 1980].

The interpretation of the results of this study are limited by the configuration of the Paleozoic eastern margin of North America. It remains a possibility that the Carolina Slate Belt region was a displaced terrain in the late Paleozoic if paleomotion in this region was east-west. Although the 'Acadian microcontinent' rocks have a counterpart to the south in the Carolina Slate belt rocks, they do not appear to share the same paleomagnetic history. The Upper Devonian to upper Carboniferous left-lateral displacement between the North American craton and the Avalon block is not detected at the southeastern margin of North America.

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